

ULAN-UDE BAIKAL, Russian Federation

WMO: 308230

Lat: 51.808N

Lon: 107.438E

Elev: 1690

StdP: 13.82

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.0	-28.8	-39.7	0.6	-33.3	-34.5	0.8	-28.3	20.8	8.3	17.5	4.1	1.1	90	0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	89.7	64.3	85.9	63.7	82.2	62.5	68.0	81.8	66.3	79.3	64.7	77.2	7.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	93.9	72.5	62.0	88.5	71.1	60.4	83.4	70.5	33.4	81.9	32.0	79.5	30.7	77.2	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.5	16.1	-37.2	96.5	5.8	3.9	-41.3	99.3	-44.7	101.5	-47.9	103.7	-52.1	106.5		
(5)			-36.9	71.4	5.6	2.0	-41.0	72.8	-44.2	73.9	-47.3	75.0	-51.4	76.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	32.2	-10.1	-1.0	19.2	37.6	50.7	64.0	68.7	63.5	49.9	33.2	12.6	-4.2	(6)
(7)		DBStd	28.52	10.65	10.37	11.65	7.83	7.75	7.32	5.65	6.02	7.08	8.10	11.20	11.46	(7)
(8)		HDD50	8129	1864	1429	956	379	87	2	0	1	87	521	1121	1682	(8)
(9)		HDD65	12259	2329	1849	1421	823	446	106	28	100	454	985	1571	2147	(9)
(10)		CDD50	1621	0	0	0	6	109	422	579	419	85	1	0	0	(10)
(11)		CDD65	275	0	0	0	0	3	76	142	53	1	0	0	0	(11)
(12)		CDH74	3498	0	0	0	6	174	1085	1526	653	54	0	0	0	(12)
(13)		CDH80	1246	0	0	0	1	44	409	586	197	9	0	0	0	(13)
(14)	Wind	WSAvg	5.4	3.0	3.4	5.2	7.5	8.1	7.1	6.2	5.6	5.8	5.1	4.2	3.4	(14)
(15)	Precipitation	PrecAvg	10.1	0.2	0.1	0.1	0.3	0.6	1.5	2.8	2.5	1.1	0.3	0.4	0.5	(15)
(16)		PrecMax	16.0	0.4	0.4	0.3	1.2	1.7	4.2	5.5	5.7	2.1	0.8	0.9	1.2	(16)
(17)		PrecMin	4.2	0.0	0.0	0.0	0.0	0.0	0.1	0.6	0.4	0.2	0.0	0.1	0.0	(17)
(18)		PrecStd	2.6	0.1	0.1	0.1	0.3	0.4	1.0	1.3	1.1	0.6	0.2	0.2	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.8	30.1	53.3	71.8	85.7	93.9	95.3	91.1	80.3	62.7	42.6	28.2	(19)
MCWB			20.4	25.5	40.3	49.7	57.8	63.7	66.9	65.4	59.4	48.3	35.1	25.2	(20)	
2%		DB	15.4	24.4	46.3	64.5	78.8	89.1	90.7	85.0	73.6	56.0	36.0	22.2	(21)	
		MCWB	13.7	21.1	36.1	46.2	54.8	62.3	65.6	64.5	57.2	44.0	31.0	20.0	(22)	
5%		DB	10.3	19.7	41.0	58.9	73.7	84.6	86.3	80.9	69.0	50.8	32.4	17.3	(23)	
		MCWB	8.9	16.9	32.9	43.6	52.8	61.6	65.2	63.3	54.9	41.3	28.6	15.6	(24)	
10%		DB	5.6	15.5	36.6	53.8	68.2	80.6	82.4	77.0	64.3	46.3	28.6	12.7	(25)	
		MCWB	4.3	13.2	30.2	40.8	50.0	60.5	64.0	62.1	52.4	39.0	25.7	11.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	25.7	40.7	50.7	59.4	67.7	71.0	69.6	62.4	49.1	35.1	25.0	(27)
MCD B			22.8	29.9	52.5	70.4	81.1	83.9	87.3	82.8	74.9	61.6	41.3	27.4	(28)	
2%		WB	13.4	20.8	36.6	47.0	56.3	65.4	68.8	66.9	58.6	44.9	31.5	20.1	(29)	
		MCD B	15.2	24.0	45.4	62.8	76.0	81.2	82.9	79.4	70.5	53.9	35.8	22.1	(30)	
5%		WB	8.7	17.1	33.2	44.1	53.6	63.6	67.3	65.1	56.0	42.0	28.8	15.6	(31)	
		MCD B	10.3	19.8	40.2	57.9	70.8	79.4	80.5	76.5	67.2	49.9	32.2	17.0	(32)	
10%		WB	4.4	13.1	30.4	41.3	51.1	61.8	65.7	63.3	53.4	39.1	25.8	11.5	(33)	
		MCD B	5.6	15.2	36.1	53.0	66.4	77.3	78.2	74.0	62.7	45.4	28.6	12.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.8	24.8	24.9	24.2	26.1	24.8	22.4	22.1	23.0	21.1	18.4	17.3	(35)
MCDBR			19.8	26.0	27.0	35.2	37.4	33.4	31.1	31.0	32.4	29.5	20.3	18.1	(36)	
MCWBR			18.2	22.7	18.0	20.1	18.2	13.2	11.7	13.5	17.2	18.5	15.9	16.5	(37)	
5% WB		MCD BR	19.6	26.1	26.1	32.8	34.0	27.6	24.3	25.2	29.1	27.6	19.6	17.9	(38)	
		MCWBR	18.1	22.8	17.7	19.5	17.0	11.9	10.5	12.1	16.0	17.7	15.4	16.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.270	0.293	0.326	0.373	0.442	0.440	0.431	0.388	0.337	0.311	0.293	0.272	(40)	
taud		2.151	2.135	2.180	2.242	2.064	2.190	2.259	2.366	2.459	2.442	2.267	2.140	(41)		
Ebn at Noon		229	258	273	273	257	259	259	266	270	255	220	204	(42)		
Edh at Noon		23	31	36	39	49	44	40	35	28	23	20	20	(43)		
(44)	All-Sky Solar Radiation	RadAvg	399	734	1179	1556	1864	1939	1794	1512	1149	717	399	279	(44)	
(45)		RadStd	22	36	56	87	104	111	106	73	105	50	29	22	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (11 neighbors)	N/A	N/A	-2.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page